YEAR 11 PHYSICS, AQA GCSE, LESSON 6 OF 10

Atoms & Nuclear Radiation.

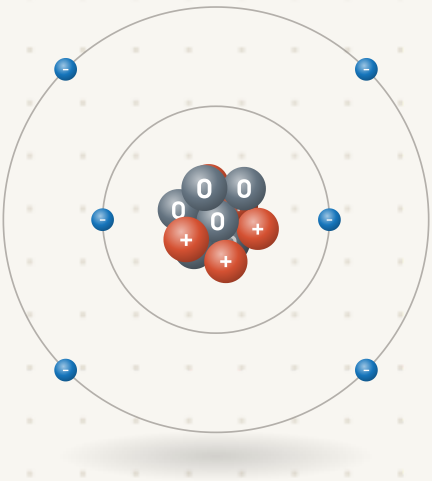
Lesson 06 Nuclear equations

Radium-226 decays and becomes radon, a completely different element. How can we predict exactly which nucleus it turns into?

IN THIS LESSON YOU WILL

- use the symbols for alpha, beta, gamma and neutron radiation
- balance a nuclear equation using mass numbers and atomic numbers
- work out the new nucleus after alpha or beta decay
- explain the effect of each decay on the mass and charge of the nucleus

ANSWER EDITION FOR TEACHERS AND SELF-MARKING



NAME _____ CLASS _____

TEACHER _____ TARGET GRADE _____

START HERE

How this booklet works.

This is the booklet for **Lesson 6** of the Atoms and Nuclear Radiation unit. It builds on the lessons before it; the "Do now" questions and the quick answers at the back help you check what you remember. Keep it: it is your classwork book *and* your revision guide.

DO NOW
Quick questions from earlier lessons. Remembering again is what makes it stick.

LEARN IT
Explanations, diagrams and worked examples. Read this again when you revise.

TRY IT
Practice that builds up: first with support, then on your own, then a challenge.

LAB
Record what you find in a simulation or a model experiment.

EXAM CORNER
AQA-style questions with a tip on how the marks are given.

REVISE IT
Summary, cover-and-answer questions and an "I can" check to colour in.

! **Revising at home?** (1) Read the Learn it pages. (2) Cover them and answer the "Cover the page" questions in Revise it. (3) Check with the quick answers at the back. (4) Go through the matching slides in the teaching deck, and try the simulations.

The route through the unit (this lesson is highlighted)

01 Inside the atom

02 Isotopes and ions

03 From plum pudding to the nucleus

04 Energy levels and the nucleus

05 Radioactive decay

06 Nuclear equations
Symbols for each radiation, balancing mass and atomic numbers (slides 28–31)

07 Properties and uses of radiation

08 Half-life and random decay

09 Half-life calculations

10 Contamination and irradiation

Teaching deck: **panphy.app/gcsephy/decks/atoms-and-radiation**. Simulations: **panphy.app/simulations/atomic_models.html** and **panphy.app/simulations/nuclear_decay.html**

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LESSON

06

Nuclear equations.

AQA Physics 4.4.2.2; Combined Science 6.4.2.2

BIG QUESTION

Radium-226 decays and becomes radon, a completely different element. How can we predict exactly which nucleus it turns into?

BY THE END YOU CAN

- use the symbols for alpha, beta, gamma and neutron radiation
- balance a nuclear equation using mass numbers and atomic numbers
- work out the new nucleus after alpha or beta decay
- explain the effect of each decay on the mass and charge of the nucleus

nuclear equation | parent nucleus | daughter nucleus | mass number | atomic number | balance

SLIDES AND SIMULATIONS

Teaching deck, slides 28–31

Alpha, beta and gamma equations, and the 'What changes?' table.

panphy.app/gcsephy/decks/atoms-and-radiation/#28

Nuclear Decay simulation

panphy.app/simulations/nuclear_decay.html

RETRIEVE

Do now

From earlier lessons. No notes!

1

What is an alpha particle made of?

2 protons and 2 neutrons (a helium nucleus).

2

What happens in the nucleus in beta decay?

A neutron turns into a proton, and a fast electron is emitted.

3

What is the unit of activity?

Becquerel (Bq).

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EXPLAIN

Balancing nuclear equations

Symbols for nuclear radiation

Emitted	Symbol	Mass number	Atomic number
Alpha particle	$^{4}_{2}\mathrm{He}$	decreases by 4	decreases by 2
Beta particle	$^{0}_{-1}\mathrm{e}$	no change	increases by 1
Gamma ray	γ	no change	no change
Neutron	$^{1}_{0}\mathrm{n}$	decreases by 1	no change

An electron has almost no mass, so its top number is 0. Its charge is −1, so its bottom number is −1. When a neutron (charge 0) turns into a proton (charge +1), the beta particle carries away the −1: the charges still balance.

! In every nuclear equation, the **top numbers add up to the same total** on both sides, and so do the **bottom numbers**. If the atomic number changes, a **new element** forms.

The three equations from the slides

$$^{241}_{95}\mathrm{Am} \rightarrow {}^{237}_{93}\mathrm{Np} + {}^{4}_{2}\mathrm{He}$$

Alpha. Top: 241 = 237 + 4; bottom: 95 = 93 + 2

$$^{14}_{6}\mathrm{C} \rightarrow {}^{14}_{7}\mathrm{N} + {}^{0}_{-1}\mathrm{e}$$

Beta. Top: 14 = 14 + 0; bottom: 6 = 7 + (−1)

$$^{99\mathrm{m}}_{43}\mathrm{Tc} \rightarrow {}^{99}_{43}\mathrm{Tc} + \gamma$$

Gamma. The ‘m’ means the nucleus has extra energy; A and Z do not change

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WORKED EXAMPLE

Radium-226, $^{226}_{88}Ra$, emits an alpha particle and becomes radon (Rn). Write the equation.

TOP	226 − 4 = 222
BOTTOM	88 − 2 = 86
NEW NUCLEUS	$^{222}_{86}Rn$
EQUATION	$^{226}_{88}Ra \rightarrow {}^{222}_{86}Rn + {}^{4}_{2}He$

YOUR TURN

Uranium-238, $^{238}_{92}U$, emits an alpha particle and becomes thorium (Th). Write the equation.

TOP	238 − 4 = 234
BOTTOM	92 − 2 = 90
NEW NUCLEUS	$^{234}_{90}Th$
EQUATION	$^{238}_{92}U \rightarrow {}^{234}_{90}Th + {}^{4}_{2}He$

WORKED EXAMPLE

Strontium-90, $^{90}_{38}Sr$, emits a beta particle and becomes yttrium (Y). Write the equation.

TOP	90 − 0 = 90
BOTTOM	38 + 1 = 39
NEW NUCLEUS	$^{90}_{39}Y$
EQUATION	$^{90}_{38}Sr \rightarrow {}^{90}_{39}Y + {}^{0}_{-1}e$

YOUR TURN

Iodine-131, $^{131}_{53}I$, emits a beta particle and becomes xenon (Xe). Write the equation.

TOP	131
BOTTOM	53 + 1 = 54
NEW NUCLEUS	$^{131}_{54}Xe$
EQUATION	$^{131}_{53}I \rightarrow {}^{131}_{54}Xe + {}^{0}_{-1}e$

PRACTISE

Try it

1 SUPPORTED Fill in the missing particle. Check that the top and bottom numbers balance.

$$^{210}_{84}Po \rightarrow {}^{206}_{82}Pb + \boxed{{}^{4}_{2}He}$$

$$^{60}_{27}Co \rightarrow {}^{60}_{28}Ni + \boxed{{}^{0}_{-1}e}$$

2 ON YOUR OWN Fill in the missing nucleus. The element symbols are given.

$$^{222}_{86}Rn \rightarrow {}^{218}_{84}Po + {}^{4}_{2}He$$

Po = polonium

$$^{137}_{55}Cs \rightarrow {}^{137}_{56}Ba + {}^{0}_{-1}e$$

Ba = barium

$$^{5}_{2}He \rightarrow {}^{4}_{2}He + {}^{1}_{0}n$$

a helium-5 nucleus emits a neutron

3 Name the type of decay in each equation.

Equation	Type of decay
$^{40}_{19}K \rightarrow {}^{40}_{20}Ca + {}^{0}_{-1}e$	beta
$^{238}_{92}U \rightarrow {}^{234}_{90}Th + {}^{4}_{2}He$	alpha
$^{99m}_{43}Tc \rightarrow {}^{99}_{43}Tc + \gamma$	gamma

4 Complete the table to show the effect of each decay on the nucleus.

Radiation emitted	Effect on the mass of the nucleus	Effect on the charge of the nucleus
alpha	decreases (mass number − 4)	decreases (by 2)
beta	no change	increases (by 1)
gamma	no change	no change

5 Explain what is wrong with Leo's statement.

Leo: In beta decay the nucleus loses an electron, so the mass number goes down by 1.

An electron has almost no mass (0 on top), so the mass number does not change. A neutron turns into a proton, so the atomic number goes up by 1.

6 CHALLENGE Uranium-238, $^{238}_{92}U$, decays by alpha emission. The new nucleus then decays by beta emission, and the next one by beta emission again. Find the final nucleus. What do you notice?

$^{238}_{92}U \rightarrow {}^{234}_{90}Th \rightarrow {}^{234}_{91}Pa \rightarrow {}^{234}_{92}U$. The final nucleus is uranium-234: an isotope of the uranium we started with (same 92 protons, 4 fewer neutrons).

AQA-STYLE

Exam corner

Write in full sentences. Show every step of a calculation.

Get the marks: Both numbers earn a mark. Check: top 212 = ? + 4, bottom 83 = ? + 2.

7 Bismuth-212, $^{212}_{83}Bi$, decays by alpha emission to form thallium (Tl). Complete the nuclear equation. [2 marks]

$$^{212}_{83}Bi \rightarrow {}^{208}_{81}Tl + {}^{4}_{2}He$$

Mass number 208 (1); atomic number 81 (1).

8 Technetium-99m emits gamma radiation. Explain why the element does not change. [2 marks]

Gamma radiation is electromagnetic radiation with no mass and no charge (1), so the number of protons / atomic number does not change (1).

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REVIEW LATER

Revise it

Use this box after the lesson and before tests.

SUMMARY

- Symbols: alpha $^{4}_{2}\mathrm{He}$, beta $^{0}_{-1}\mathrm{e}$, gamma γ, neutron $^{1}_{0}\mathrm{n}$.
- Top numbers balance; bottom numbers balance.
- Alpha: mass number − 4, atomic number − 2.
- Beta: mass number unchanged, atomic number + 1.
- Gamma: no change to either number.

COVER THE PAGE AND ANSWER

1 What is the symbol for a beta particle?

2 What happens to the atomic number in alpha decay?

3 Which decay does not change the mass number or atomic number?

4 Complete: $^{226}_{88}\mathrm{Ra} \rightarrow ? + {}^{4}_{2}\mathrm{He}$

I CAN…

- write the symbols for alpha, beta, gamma and neutron radiation
- balance a nuclear equation
- find the new nucleus after alpha or beta decay
- explain the effect of each decay on mass and charge

REFERENCE

Toolkit.

The facts and methods you have met so far. Learn these: exam questions expect you to recall them without a data sheet.

The three particles

Lesson 1

Particle	Charge	Mass	Where
Proton	+1	1	nucleus
Neutron	0	1	nucleus
Electron	−1	very small	energy levels

Reading the symbol

Lesson 1

$^{A}_{Z}X$ e.g. $^{23}_{11}Na$

- Z, atomic number = protons
- A, mass number = protons + neutrons
- neutrons = A − Z; in an atom, electrons = Z
- Atom radius ≈ 1×10^{-10} m; nucleus radius < 1/10 000 of that

Isotopes and ions

Lesson 2

- **Isotopes**: same number of protons, different numbers of neutrons.
- **Positive ion**: an atom that has lost one or more outer electrons. The nucleus is unchanged.

Alpha scattering

Lesson 3

Observation	Conclusion
most straight through	mostly empty space
some deflected	positive centre
a very few bounced back	mass in a tiny centre

Models and energy levels

Lesson 4

- 1897 electron → 1904 plum pudding → 1911 nuclear → 1913 Bohr → 1932 neutron (Chadwick)
- **Absorb** EM radiation → higher level, further out.
- **Emit** EM radiation → lower level, closer in.

Nuclear radiation

Lesson 5

- Unstable nuclei decay at **random** to become more stable.
- **Activity**: decays per second, in becquerel (Bq). **Count rate**: decays recorded per second by a detector.
- α: 2p + 2n; β: fast electron, a neutron → proton; γ: EM radiation; n: a neutron

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Nuclear equations

Lesson 6

Emitted	Symbol	Mass no.	Atomic no.
alpha	$^{4}_{2}\mathrm{He}$	− 4	− 2
beta	$^{0}_{-1}\mathrm{e}$	no change	+ 1
gamma	γ	no change	no change
neutron	$^{1}_{0}\mathrm{n}$	− 1	no change

Top numbers balance; bottom numbers balance.

$$^{226}_{88}\mathrm{Ra} \rightarrow {}^{222}_{86}\mathrm{Rn} + {}^{4}_{2}\mathrm{He}$$

$$^{14}_{6}\mathrm{C} \rightarrow {}^{14}_{7}\mathrm{N} + {}^{0}_{-1}\mathrm{e}$$

REFERENCE

Key words from this lesson.

Learn the meaning and spelling. In exams, precise words earn marks: say “nucleus”, not “middle”, and “unstable nucleus”, not “unstable atom”.

Daughter nucleus: The new nucleus formed when a nucleus decays.

Nuclear equation: An equation showing a decay, in which the mass numbers and atomic numbers balance.

CHECK YOURSELF

Quick answers.

Answers to the “Do now” and “Cover the page” questions, so you can check yourself when revising. Answers to the longer questions are in the answer edition.

DO NOW

1. 2 protons and 2 neutrons (a helium nucleus).
2. A neutron turns into a proton, and a fast electron is emitted.
3. Becquerel (Bq).

COVER THE PAGE

1. $^{0}_{-1}\mathrm{e}$
2. It decreases by 2.
3. Gamma emission.
4. $^{222}_{86}\mathrm{Rn}$

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